Supervyse OPF Intel Tech Pack PFR Porting Guide

Revision 1.0

06/2023

Copyright (c) 2024, All Rights Reserved. Insyde Software.

No part of this publication may be reproduced, transmitted, transcribed, stored in a retrieval system, or translated into any language or computer language, in any form, or by any means, electronic, mechanical, magnetic, optical, chemical, manual or otherwise, without the prior written permission of Insyde Software Corp.

Disclaimer

Insyde Software provides this document and the programs "as is" without warranty of any kind, either expressed or implied, including, but not limited to, the implied warranties of merchantability or fitness for a particular purpose.

This document could contain technical inaccuracies or typographical errors. Changes are periodically made to the information herein; these changes will be incorporated in future revisions of this document. Insyde Software is under no obligation to notify any person of the changes.

The following trademarks are used in this document:

Insyde and InsydeH2O are registered trademarks or trademarks of Insyde Software. All other trademarks or trade names are property of their respective holders.

Contents

1 AST2600 porting & configurations 6
1.1 Introduction 6
1.2 BMC PFR image layout 6
1.3 BMC PFR image configuration 7
1.3.1 Set pfr layout file 7
1.3.2 Set json file 8
1.3.3 PFR parameters configuration 14
1.3.4 Generating root keys and code signing keys 14
1.3.5 Installing keys 14
1.3.6 Configurable I2C bus and slave address for PFR Host Mailbox 15

Revision History

Revision Number	Date	Author	Reviewer	Summary
1.0	2023-06-14	Allen Liu	Edwin Liu	Initial Release

1 AST2600 porting & configurations

1.1 Introduction

In this chapter, we will describe how to port the BMC PFR layout and its related configurations. For more details about Intel PFR, please refer to **Intel PFR Architecture and Hardware Design Guide.**

1.2 BMC PFR image layout

- The BMC PFR image consists of one **256MB** binary which primarily includes **active**, **recovery**, **staging** region, and **Platform Firmware Manifest (PFM)**

(describing active platform firmware) and the partition layout of the BMC PFR image is shown below

Contents	Start Address	Size(Hex)	Static/Dynamic
uboot	0x0	0x0010_0000 (1024KB)	static
fit-image-a	0x0010_0000	0x0368_0000 (56320KB)	static
sofs	0x0378_0000	0x0020_0000 (2048KB)	Dynamic
rwfs	0x0398_0000	0x005e_0000 (6016KB)	Dynamic
cerificate	0x03f6_0000	0x0001_0000 (64KB)	Dynamic
sdr	0x03f7_0000	0x0001_0000 (64KB)	Dynamic
u-boot-env	0x03f8_0000	0x0002_0000 (128KB)	Dynamic
reserved1	0x03fa_0000	0x0a4_0000(10496KB)	Dynamic
pfm	0x049e_0000	0x0002_0000 (128KB)	static
rc-image	0x04a0_0000	0x0400_0000(64MB)	static
image-stg	0x08a0_0000	0x0400_0000 (64MB)	Dynamic
pch-stg	0x0ca0_0000	0x0100_0000 (16MB)	Dynamic
pfr-stg	0x0da0_0000	0x0040_0000 (4MB)	Dynamic
afm-active	0x0de0_0000	0x0002_0000 (128KB)	Dynamic
afm-recovery	0x0de2_0000	0x0002_0000 (128KB)	Dynamic

Contents	Start Address	Size(Hex)	Static/Dynamic
reserved2	0x0de4_0000	0x000c_0000 (768KB)	Dynamic
cpld-gold	0x0df0_0000	0x0010_0000 (1024KB)	Dynamic
reserved3	0x0e00_0000	0x0200_0000 (2048KB)	Dynamic

1.3 BMC PFR image configuration

- Set up mtdparts and JSON files according to BMC PFR image layout

1.3.1 Set pfr layout file

- Flash layout file is located at /meta-insyde/meta-intel/recipes-pfr/conf/machine/include/pfr-intel-flashlayout-256MB.inc file

```
FLASH_SPL_OFFSET:flash-262144        = "0"
FLASH_SPL_OFFSET[name]               = "spl"

FLASH_UBOOT_OFFSET:flash-262144      = "64"
FLASH_UBOOT_OFFSET[name]             = "u-boot"

FLASH_KERNEL_ROFS_OFFSET:flash-262144 = "1024"
FLASH_KERNEL_ROFS_OFFSET[name]       = "image-a"
FLASH_KERNEL_ROFS_OFFSET[kentry]     = "1"

FLASH_SOFS_OFFSET:flash-262144       = "56832"
FLASH_SOFS_OFFSET[name]              = "sofs"

FLASH_RWFS_OFFSET:flash-262144       = "58880"
FLASH_RWFS_OFFSET[name]              = "rwfs"

FLASH_CERTIFICATE_OFFSET:flash-262144  = "64896"
FLASH_CERTIFICATE_OFFSET[name]       = "certificate"

FLASH_SDR_OFFSET:flash-262144        = "64960"
FLASH_SDR_OFFSET[name]               = "sdr"

FLASH_UBOOT_ENV_OFFSET:flash-262144    = "65024"
FLASH_UBOOT_ENV_OFFSET[name]         = "u-boot-env"
FLASH_UBOOT_ENV_OFFSET[ubootenv]     = "1"
FLASH_UBOOT_ENV_OFFSET[dual]         = "1"

FLASH_RESERVED1_OFFSET:flash-262144    = "65152"
FLASH_RESERVED1_OFFSET[name]         = "reserved1"

FLASH_PFM_OFFSET:flash-262144        = "75648"
FLASH_PFM_OFFSET[name]               = "pfm"

FLASH_RC_IMAGE_OFFSET:flash-262144     = "75776"
FLASH_RC_IMAGE_OFFSET[name]          = "rc-image"

FLASH_IMAGE_STG_OFFSET:flash-262144    = "141312"
```

```
FLASH_IMAGE_STG_OFFSET[name]          = "image-stg"

FLASH_PCH_STG_OFFSET:flash-262144      = "206848"
FLASH_PCH_STG_OFFSET[name]            = "pch-stg"

FLASH_PFR_STG_OFFSET:flash-262144      = "223232"
FLASH_PFR_STG_OFFSET[name]            = "pfr-stg"

FLASH_AFM_ACTIVE_OFFSET:flash-262144   = "227328"
FLASH_AFM_ACTIVE_OFFSET[name]          = "afm-active"

FLASH_AFM_RC_OFFSET:flash-262144       = "227456"
FLASH_AFM_RC_OFFSET[name]             = "afm-recovery"

FLASH_RESERVED2_OFFSET:flash-262144    = "227584"
FLASH_RESERVED2_OFFSET[name]          = "reserved2"

FLASH_CPLD_GOLD_OFFSET:flash-262144    = "228352"
FLASH_CPLD_GOLD_OFFSET[name]          = "cpld-gold"

FLASH_RESERVED3_OFFSET:flash-262144    = "229376"
FLASH_RESERVED3_OFFSET[name]          = "reserved3"

PFR_MANIFEST:flash-262144 = "pfr_manifest_intel_pfr_256.json"

FLASH_SEQ = "FLASH_SPL_OFFSET FLASH_UBOOT_OFFSET FLASH_KERNEL_ROFS_OFFSET
FLASH_SOFS_OFFSET FLASH_RWFS_OFFSET FLASH_CERTIFICATE_OFFSET FLASH_SDR_OFFSET
FLASH_UBOOT_ENV_OFFSET FLASH_RESERVED1_OFFSET FLASH_PFM_OFFSET FLASH_RC_IMAGE_OFFSET
FLASH_IMAGE_STG_OFFSET FLASH_PCH_STG_OFFSET FLASH_PFR_STG_OFFSET
FLASH_AFM_ACTIVE_OFFSET FLASH_AFM_RC_OFFSET FLASH_RESERVED2_OFFSET
FLASH_CPLD_GOLD_OFFSET FLASH_RESERVED3_OFFSET"
```

- According to pfr-intel-flashlayout-256MB.inc file, uboot recipe will automatically generate mtdparts parameter of the kernel command line. Users can confirm whether it is consistent with the conf file.

```
mtdparts=bmc:64k@0x0(spl),960k@0x10000(u-boot),55808k@0x100000(image-
a),2048k@0x3780000(sofs),6016k@0x3980000(rwfs),64k@0x3f60000(certificate),64k@0x3f70000(sdr),128k@0x3f8
0000(u-boot-env),10496k@0x3fa0000(reserved1),128k@0x49e0000(pfm),65536k@0x4a00000(rc-
image),65536k@0x8a00000(image-stg),16384k@0xca00000(pch-stg),4096k@0xda00000(pfr-stg),128k@0xde00000(afm-
active),
128k@0xde20000(afm-recovery),768k@0xde40000(reserved2),1024k@0xdf00000(cpld-gold),32768k@0xe000000(reserved3)
```

1.3.2 Set json file

- PFR json file is located at meta-insyde/meta-intel/recipes-pfr/intel-pfr/obmc-intel-pfr-image-native/pfr_manifest_intel_pfr_256.json file

```
{
   "exclude-pages":[[18912,18943],[18944,65535]],
   "image-parts": [{
         "name": "u-boot",
         "index": 0,
         "offset": "0",
         "size": "0x100000",
         "prot_mask": 29,
         "pfm": 1,
         "hash": 1,
         "compress": 1
      },
      {
         "name": "fit-image-a",
```

```
        "index": 1,
        "offset": "0x100000",
        "size": "0x3680000 ",
        "prot_mask": 29,
        "pfm": 1,
        "hash": 1,
        "compress": 1
    },
    {
        "name": "sofs",
        "index": 2,
        "offset": "0x3780000",
        "size": "0x200000",
        "prot_mask": 31,
        "pfm": 1,
        "hash": 0,
        "compress": 1
    },
    {
        "name": "rwfs",
        "index": 3,
        "offset": "0x3980000",
        "size": "0x5e0000 ",
        "prot_mask": 31,
        "pfm": 1,
        "hash": 0,
        "compress": 1
    },
    {
        "name": "cerificate",
        "index": 4,
        "offset": "0x3f60000",
        "size": "0x10000 ",
        "prot_mask": 31,
        "pfm": 1,
        "hash": 0,
        "compress": 1
    },
    {
        "name": "sdr",
        "index": 5,
        "offset": "0x3f70000",
        "size": "0x10000",
        "prot_mask": 31,
        "pfm": 1,
        "hash": 0,
        "compress": 1
    },
    {
        "name": "u-boot-env",
        "index": 6,
        "offset": "0x03f80000",
        "size": "0x20000",
        "prot_mask": 31,
        "pfm": 1,
        "hash": 0,
        "compress": 1
    },
    {
        "name": "reserved1",
        "index": 7,
```

```
        "offset": "0x3fa0000",
        "size": "0xa40000",
        "prot_mask": 31,
        "pfm": 1,
        "hash": 0,
        "compress": 0
    },
    {
        "name": "pfm",
        "index": 8,
        "offset": "0x49e0000",
        "size": "0x20000",
        "prot_mask": 1,
        "pfm": 1,
        "hash": 0,
        "compress": 0
    },
    {
        "name": "rc-image",
        "index": 9,
        "offset": "0x4a00000",
        "size": "0x4000000",
        "prot_mask": 0,
        "pfm": 1,
        "hash": 0,
        "compress": 0
    },
    {
        "name": "image-stg",
        "index": 10,
        "offset": "0x8a00000",
        "size": "0x5400000",
        "prot_mask": 3,
        "pfm": 1,
        "hash": 0,
        "compress": 0
    },
    {
        "name": "afm-active",
        "index": 11,
        "offset": "0xde00000",
        "size": "0x20000",
        "prot_mask": 0,
        "pfm": 1,
        "hash": 0,
        "compress": 0
    },
    {
        "name": "afm-recovery",
        "index": 12,
        "offset": "0xde20000",
        "size": "0x20000",
        "prot_mask": 0,
        "pfm": 1,
        "hash": 0,
        "compress": 0
    },
    {
        "name": "reserved2",
        "index": 13,
        "offset": "0xde40000",
```

```
        "size": "0xc0000",
        "prot_mask": 0,
        "pfm": 1,
        "hash": 0,
        "compress": 0
      },
      {
        "name": "cpld-gold",
        "index": 14,
        "offset": "0xdf00000",
        "size": "0x100000",
        "prot_mask": 0,
        "pfm": 1,
        "hash": 0,
        "compress": 0
      }
    ],
    "i2c-rules": [{
        "bus-id": 3,
        "rule-id": 3,
        "address": "0xD0",
        "cmd-whitelist": ["0x00", "0x01", "0x02", "0x03", "0x04", "0x09", "0x0A", "0x0B", "0x0C", "0x0D", "0x0E", "0x0F",
                          "0x10", "0x13", "0x17", "0x1B", "0x1C", "0x1D", "0x02", "0x021", "0x22", "0x23", "0x25", "0x30",
                          "0x31", "0x32", "0x33", "0x035", "0x36", "0x37", "0x38", "0x39", "0x3A", "0x3B", "0x3C", "0x3D"]
          },
          {
        "bus-id": 3,
        "rule-id": 4,
        "address": "0xD8",
        "cmd-whitelist": ["0x00", "0x01", "0x02", "0x03", "0x04", "0x09", "0x0A", "0x0B", "0x0C", "0x0D", "0x0E", "0x0F",
                          "0x10", "0x13", "0x17", "0x1B", "0x1C", "0x1D", "0x02", "0x021", "0x22", "0x23", "0x25", "0x30",
                          "0x31", "0x32", "0x33", "0x035", "0x36", "0x37", "0x38", "0x39", "0x3A", "0x3B", "0x3C", "0x3D"]
      },
      {
        "bus-id": 1,
        "rule-id": 6,
        "address": "0xB0",
        "cmd-whitelist": ["0x00", "0x03", "0x05", "0x06", "0x19", "0x1A", "0x30", "0x3A", "0x3B", "0x3C", "0x3D", "0x3E",
"0x3F",
                          "0x79", "0x7A", "0x7B", "0x7C", "0x7D", "0x7E", "0x7F", "0x81", "0x82", "0x86", "0x87", "0x88",
                          "0x89", "0x8C", "0x8D", "0x8E", "0x8F", "0x90", "0x91", "0x92", "0x93", "0x94", "0x95", "0x96",
                          "0x97", "0x98", "0x9A", "0xA6", "0xA7", "0xD0", "0xD3", "0xD4", "0xD5", "0xD6", "0xD7", "0xD8",
                          "0xD9", "0xDC", "0xDD", "0xDE", "0xDE"]
      },
      {
        "bus-id": 1,
        "rule-id": 4,
        "address": "0xB2",
        "cmd-whitelist": ["0x00", "0x03", "0x05", "0x06", "0x19", "0x1A", "0x30", "0x3A", "0x3B", "0x3C", "0x3D", "0x3E",
"0x3F",
                          "0x79", "0x7A", "0x7B", "0x7C", "0x7D", "0x7E", "0x7F", "0x81", "0x82", "0x86", "0x87", "0x88",
                          "0x89", "0x8C", "0x8D", "0x8E", "0x8F", "0x90", "0x91", "0x92", "0x93", "0x94", "0x95", "0x96",
                          "0x97", "0x98", "0x9A", "0xA6", "0xA7", "0xD0", "0xD3", "0xD4", "0xD5", "0xD6", "0xD7", "0xD8",
                          "0xD9", "0xDC", "0xDD", "0xDE", "0xDE"]
      },
      {
        "bus-id": 2,
        "rule-id": 1,
        "address": "0xB4",
        "cmd-whitelist": ["0x00"]
      },
```

```
    {
      "bus-id": 2,
      "rule-id": 2,
      "address": "0xD4",
      "cmd-whitelist": ["0x00"]
    },
    {
      "bus-id": 2,
      "rule-id": 3,
      "address": "0x4A",
      "cmd-whitelist": ["0x00"]
    },
    {
      "bus-id": 2,
      "rule-id": 4,
      "address": "0x4C",
      "cmd-whitelist": ["0x00"]
    },
    {
      "bus-id": 2,
      "rule-id": 5,
      "address": "0xDC",
      "cmd-whitelist": ["0x00"]
    },
    {
      "bus-id": 2,
      "rule-id": 6,
      "address": "0xEC",
      "cmd-whitelist": ["0x00"]
    },
    {
      "bus-id": 2,
      "rule-id": 7,
      "address": "0xE0",
      "cmd-whitelist": ["0x00"]
    },
    {
      "bus-id": 2,
      "rule-id": 8,
      "address": "0xB0",
      "cmd-whitelist": ["0x00"]
    },
    {
      "bus-id": 2,
      "rule-id": 9,
      "address": "0xC4",
      "cmd-whitelist": ["0x00"]
    },
    {
      "bus-id": 2,
      "rule-id": 10,
      "address": "0xCC",
      "cmd-whitelist": ["0x00"]
    },
    {
      "bus-id": 2,
      "rule-id": 11,
      "address": "0xE4",
      "cmd-whitelist": ["0x00"]
    }
```

```
  ]
}
```

- Partition Protection Information Definition

Property	Description
name	Partition label defined in dts
index	Partition index
offset	Start offset of the partition
size	Size of the partition
prot_mask	• Bit[0]: Protect: Read allowed 1 • Bit[1]: Protect: Write allowed 1; 0 - not allowed • Bit[2]: Recover: Recover on first recovery 1 • Bit[3]: Recover: Recover on second recovery 1 • Bit[4]: Recover: Recover on third recovery 1 • Bit[7:5]: Reserved
pfm	Set to 1 if the partition information should be added in PFM.
hash	Set to 1 if hashing of the region is needed.
compress	Set to 1 if the partition is compressed in the capsule

- Example

```
{
  "name": "rwfs",
  "index": 3,
  "offset": "0x3980000",
  "size": "0x5e0000 ",
  "prot_mask": 31,
  "pfm": 1,
  "hash": 0,
  "compress": 1
},
```

- I2C Rules Definition

Property	Description
bus-id	I2C bus number
rule-id	Rule id of the bus
address	I2C device address
cmd-whitelist	Allowed i2c commands

- Example

```
{
  "bus-id": 1,
  "rule-id": 4,
  "address": "0xB2",
```

```
   "cmd-whitelist": ["0x00", "0x03", "0x05", "0x06", "0x19", "0x1A", "0x30", "0x3A", "0x3B", "0x3C", "0x3D", "0x3E",
"0x3F",
                "0x79", "0x7A", "0x7B", "0x7C", "0x7D", "0x7E", "0x7F", "0x81", "0x82", "0x86", "0x87", "0x88",
                "0x89", "0x8C", "0x8D", "0x8E", "0x8F", "0x90", "0x91", "0x92", "0x93", "0x94", "0x95", "0x96",
                "0x97", "0x98", "0x9A", "0xA6", "0xA7", "0xD0", "0xD3", "0xD4", "0xD5", "0xD6", "0xD7", "0xD8",
                "0xD9", "0xDC", "0xDD", "0xDE", "0xDE"]
},
```

1.3.3 PFR parameters configuration

- Set meta-insyde/meta-intel/recipes-pfr/images/intel-platforms.bbappend file

Parameters	Description	Default Value
PFR_SVN	PFR image security version number, range is 0 – 64 SVN is checked for update. No update action if SVN is smaller.	1
PFR_BKC_VER	BKC version.	1
PFR_BUILD_VER_MAJ	PFR image build major version, range is 0x00 – 0xff	1
PFR_BUILD_VER_MIN	PFR image build minor version, range is 0x00 – 0xff	0
PFR_BUILD_NUM	OEM specific data, range is 0x000000 – 0xffffff	0x565566

1.3.4 Generating root keys and code signing keys

Intel PFR supports scep256r1 and scep384r1 signing algorithm, please run the following commands in your host for generating ECDSA keys.

- **SHA256**

```
# openssl ecparam -name secp256r1 -genkey -out rk_prv.pem
# openssl ec -in rk_prv.pem -pubout -out rk_pub.pem
# openssl ecparam -name secp256r1 -genkey -out csk_prv.pem
# openssl ec -in csk_prv.pem -pubout -out csk_pub.pem
```

- **SHA384**

```
# openssl ecparam -name secp384r1 -genkey -out rk384_prv.pem
# openssl ec -in rk384_prv.pem -pubout -out rk384_pub.pem
# openssl ecparam -name secp384r1 -genkey -out csk384_prv.pem
# openssl ec -in csk384_prv.pem -pubout -out csk384_pub.pem
```

1.3.5 Installing keys

- The generated ECDSA keys should be put in meta-openbmc-mods/meta-egs/recipes-intel/intel-pfr/obmc-intel-pfr-image-native to override it.
- File name should be:
    - rk_prv.pem, rk_pub.pem, csk_prv.pem, and csk_pub.pem for secp256r1 keys.

- rk384_prv.pem, rk384_pub.pem, csk384_prv.pem, and csk384_pub.pem for secp384r1 keys.

1.3.6 Configurable I2C bus and slave address for PFR Host Mailbox

- Set /meta-insyde/meta-intel/recipes-phosphor/fw-update/files/fwupd-restricted.sh file

```
PFR_BUS=4
PFR_ADDR=0x38
```